# fn make_privacy_filter

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `fn make_privacy_filter`.

## 1 Hoare Triple

### Precondition

#### Compiler-verified

- Argument `odometer` of type `Odometer<DI, MI, MO, Q, A>`.
- Argument `d_in` of type `MI_Distance`, the associated distance type of `MI`.
- Argument `d_out` of type `MO_Distance`, the associated distance type of `MO`.
- Generic `DI` implements `Domain`.
- Generic `MI` implements `Metric`.
- Generic `MO` implements `Measure`.
- `MI_Distance` implements `ProductOrd`.
- `MO_Distance` implements `ProductOrd`.
- `(DI, MI)` implements `MetricSpace`.

#### User-verified

None

### Pseudocode

```python
def make_privacy_filter(
    odometer: Odometer[DI, MI, MO, Q, A],
    d_in: MI_Distance,
    d_out: MO_Distance,
) -> Measurement[DI, MI, MO, OdometerQueryable[MI, MO, Q, A]]:
    odo_function = odometer.function
    continuation_rule = new_continuation_rule(  #
        d_in, d_out, Q, A, MI_Distance, MO_Distance
    )

    def function(arg: DI_Carrier) -> OdometerQueryable[MI, MO, Q, A]:
        return wrap(continuation_rule, lambda: odo_function.eval(arg))  #

```

```
14     def privacy_map(d_in_p: MI_Distance) -> MO_Distance:
15         if d_in_p.total_gt(d_in):
16             raise "input distance must not be greater than d_in"
17 
18         return d_out
19 
20     return Measurement.new(
21         odometer.input_domain,
22         odometer.input_metric,
23         odometer.output_measure,
24         Function.new_interactive(function),
25         PrivacyMap.new_fallible(privacy_map),
26     )
```

**Postcondition**

For every setting of the input parameters `(odometer, d_out, DI, MI, MO, Q, A)` to `make_privacy_filter` such that the given preconditions hold, `make_privacy_filter` raises an error (at compile time or run time) or returns a valid odometer. A valid odometer has the following properties:

1. (Data-independent runtime errors). For every pair of members $x$ and $x'$ in `input_domain`, `invoke`($x$) and `invoke`($x'$) either both return the same error or neither return an error.

2. (Valid odometer queryable). For every member $x$ in `input_domain`, where `function`($x$) does not raise an error, `function`($x$) returns a valid odometer queryable.

*Proof of data-independent errors.* `Function.eval` on line 12 has data-independent exceptions, because the function is from `odometer`, which is a valid odometer. Since this is the only location where an exception can be raised, the data-independent errors property holds. □

*Proof of privacy guarantee.* By the definition of a valid odometer queryable, the output of `make_privacy_filter` upholds the privacy guarantee. □